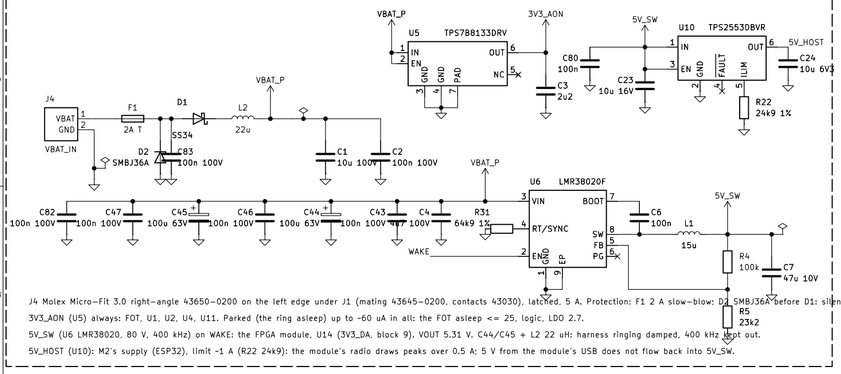
1. POWER J4 battery -> 3V3_AON (always) and 5V_SW (on WAKE) -> 5V_HOST (M2)



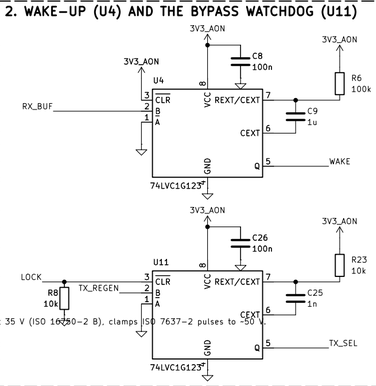
J4 Molex Micro-Fit 3.0 right-angle 43650-0200 on the left edge under J1 (mating 43645-0200, contacts 43030), latched. 5 A. Protection: F1 2 A slow-blow; D2 SMBJ36A before D1: silent at 35 V (ISO 16750-2 B), clamps ISO 7637-2 pulses to ~90 V.

3V3_AON (U5) always: FOT, U1, U2, U4, U11. Parked (the ring asleep) up to ~60 uA in all; the FOT asleep <= 25, logic, LDO 2.7.

5V_SW (U6 LMR38020, 80 V, 400 kHz) on WAKE; the FPGA module, U14 (3V3_DA, block 9), VOUT 5.31 V. C44/C45 + L2 22 uH: harness ringing damped, 400 kHz kept out.

5V_HOST (U10): M2's supply (ESP32), limit ~1 A (R22 24k9): the module's radio draws peaks over 0.5 A; 5 V from the module's USB does not flow back into 5V_SW.

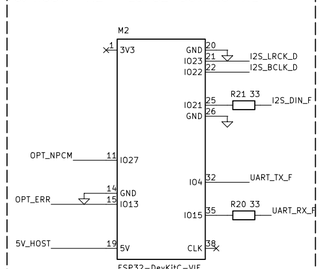
2. WAKE-UP (U4) AND THE BYPASS WATCHDOG (U11)



U4: every RX edge restarts ~0.1 s (R6 x C9). The ring asleep -> WAKE = 0 -> U6 off, the FPGA unpowered. STATUS's polarity need not be known.

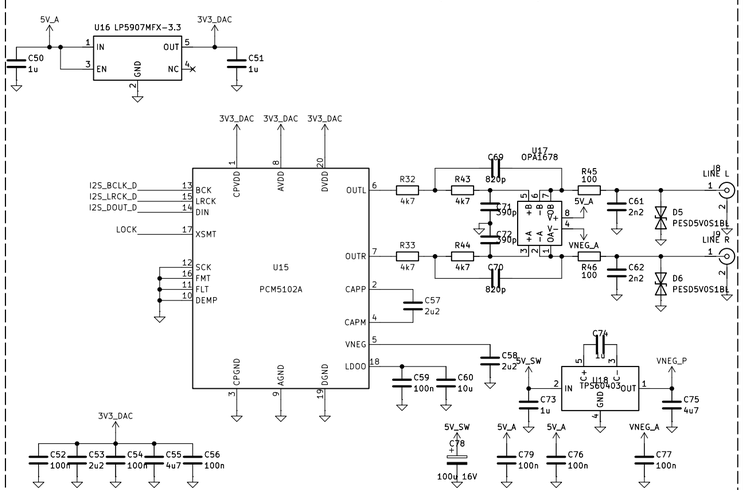
U11: TX_SEL = 1 only while LOCK = 1 and edges on TX_REGEN (R23 x C25 ~10 us; TX without edges longer is not MOST). TX stuck (clock, U9, FPGA) -> TX_SEL = 0: bypass. RB: FPGA not configured -> LOCK = 0 -> bypass.

8. HOST M2 = ESP32-DevKitC-VIE



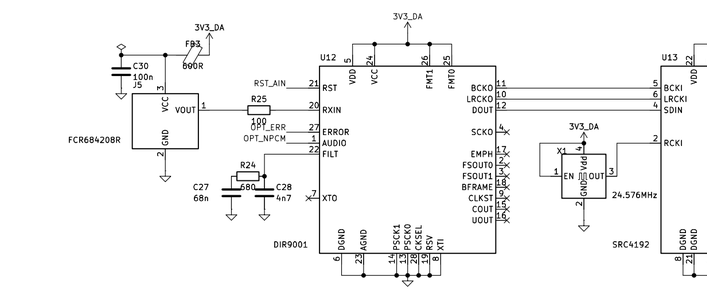
M2: ESP32-DevKitC-VIE (WROVER-IE): 8 MB PSRAM, an IPEX antenna (external, 2.4 GHz), -40...85 °C. Pins 2 x 1x19, 2.54 pitch, the rows 25.40 mm apart.
5V_HOST -> 5V; the module's USB through its own diode.
UART 3 Mbaud: IO4 <- UART_TX, IO15 -> R20 -> UART_RX.
I2S slave: IO22/23 <- BCLK/LRCK, IO21 -> R21 -> DIN.
IO13 <- OPT_ERR, IO27 <- OPT_NPCM (U12, block 9; v1.3).
v1.2: the links to the FPGA on the row facing M1.

11. LINE OUT J3's stereo -> U15 PCM5102A -> J8 L, J9 R (RCA, 2.1 Vrms)



U15 PCM5102A, hardware mode: FMT = 0 (I2S), FLT = 0 (normal latency), DEMP = 0, SCK to ground: the PLL takes its clock from BCK (3 wires, BCK = 64 Fs).
Its input is the same stereo as J3's. XSMT = LOCK: the DAC is muted while the ring is not locked and before the FPGA is configured (R8): it ramps over 104 samples.
Output 2.1 Vrms ground-referenced, no coupling caps. U17: buffer with a 2nd-order Sallen-Key low-pass (Butterworth, 60 kHz). C0G; then 100 ohm + 2.2 nF, D5, D6 (ESD) at the
U17 on +5V_SW and VNEG_A (-5.2 V, U18 TPS60403): output to the rails, input to V-, so +-3 V of signal fits with margin. J8, J9 pad 3: as J7's, measure a sample.
U16 LP5907: 3V3_DAC from 5V_SW. Its output takes at most 10 uF, so AVDD 4.7 uF and CPVDD 2.2 uF instead of fig. 33's 10 uF.
On the board each capacitor of the row stands by its pin: C52 C53 CPVDD; C54 C55 AVDD; C56 DVDD.

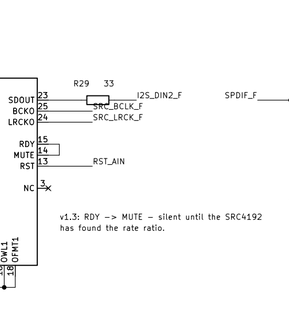
3. THE RING J1 = MOST in and out (both fibres); regeneration, always powered



fibre FROM the previous node -> RX
fibre TO the next node <- TX
R3 || R42 = 2 x 27k = 13k5: RGAIN -> TXVCC (AN 5344: 13.5..33k).
FB1/FB2 + 100n + 10u: RX and TX separately.
R7: RXD held at 0 while the receiver sleeps.

S = TX_SEL = 0: FPGA off, not locked, or TX stuck -> straight through RX -> TX.
S = 1 (LOCK and TX edges, U11): TX = the regenerated signal from the FPGA. The network does not see us.

4. FPGA ISOLATION, TX FLIP-FLOP

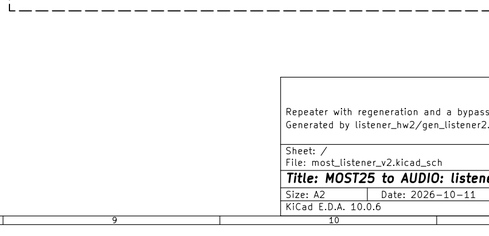


U3, U9 on 3V3_SW: when the FPGA is off, they (loff) do not load the lines.
U9: the TX edges on the SI514's clock (RT_CLK).
Q -> U2 (TX_REGEN) and into the FPGA (TX_FB).

5. FPGA Sipeed Tang Nano 20K

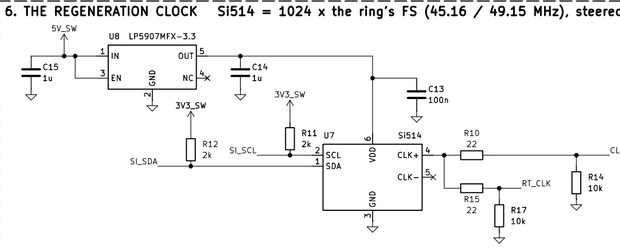
Pins 1..20 / 21..40 top to bottom, from the USB-C; name = the GW2AR pin + what the module has on it (data sheet v1.3).
The ring and the SI514 on the LCD pins only (an empty connector): LOCK on 31, S/PDIF on 42. HSPI, 54..56 (the module amplifier's inputs: PA_EN = 51 keeps it off), EDID 52/53: UART, MCLK, I2S, SD1, DIN2.
CLK_1024 (SI514) -> 77 (GCLKT_1). TX_FB -> 25 through IDES8 (1.26 ns); it takes the pair, so 26 is not connected.
D3: the USB and 5V_SW are never joined: U6 gives 5.3 V, the module gets ~5.0 V.

12. v2.0: U17'S SUPPLY FILTERS, I2S TERMINATION TO THE DAC, RX_IN



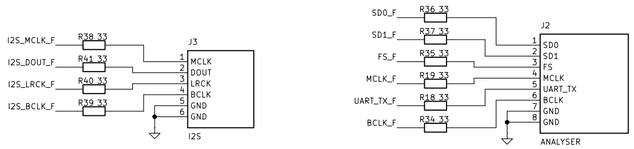
U17 (OPA1678) no longer runs straight off the rails: V+ = 5V_A (5V_SW via R47 10 ohm, C84 10 uF; it feeds U16, the DAC's LDO, too), V- = VNEG_A (U18's VNEG_P via R48 10 ohm
RC 10 ohm / 10 uF: corner 1.6 kHz, -45 dB at 250 kHz (the pump) and the buck's frequency; R47 drops up to 0.2 V (U17 + U16 with the DAC), R48 -50 mV. A bead with 10 uF wo
R49-R51: 33 ohm at the FPGA on the I2S branch to the DAC and ESP32 (BCLK, LRCK, DOUT: 35-45 mm, many loads; the DAC clocks off BCLK). J3: R39-R41.
R52: 33 ohm at U3's output (RX_IN to the FPGA, about 60 mm).
R53: 33 ohm at U1's output (RX_BUF to U2, U3, U4; through U2 and R2 the ring's TX), as R43 in the reader v2.0: drawn in block 3.

6. THE REGENERATION CLOCK SI514 = 1024 x the ring's FS (45.16 / 49.15 MHz), steered by the FPGA



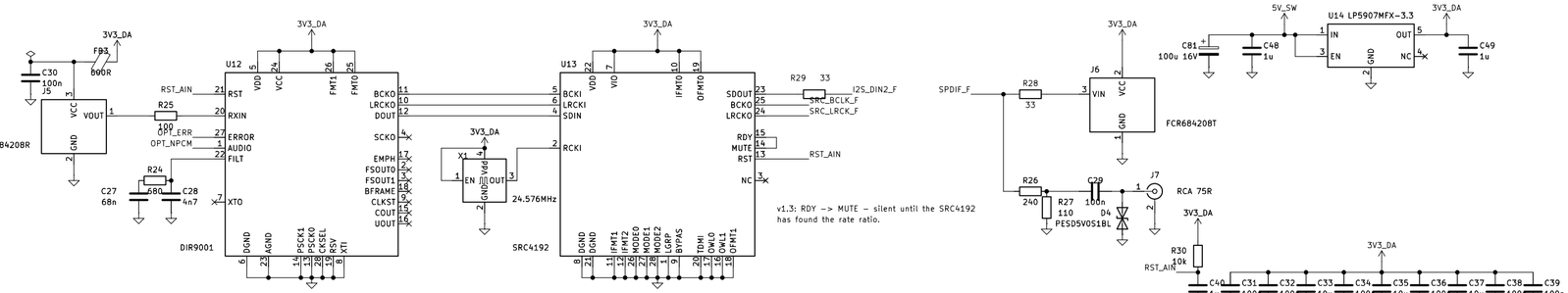
U7 SI514: any off-the-shelf 3.3 V CMOS one - after 514 in the part number the letter C, M or N; package 3.2x5 or 2.5x3.2 (the letter before AG: B or C), both fit one land pattern.
CLK+ -> FPGA (clk_r = 1024 Fs) and through R15 to U9: the TX edges sit on the SI514's clock, not on the FPGA's output. CLK- is not needed.
The FPGA finds the SI514's I2C address itself (it tries 0x08..0x77) and steers M (dpll.v): TX_FB's delay against RX_IN within 1.26 ns, steps of 0.026 ppb.
At start-up, centring (FCAL, up to 10 ms): the SI514's output is high-impedance then, R14 and R17 hold the lines at 0. U8: its own LDO from 5V_SW.

7. OUTPUTS J2: analyser (MCLK, 2 x TDM8, UART_TX); J3: stereo I2S for a DAC / DSP



J2 SDO/SD1: the frame's bytes 0..31 / 32..63, a TDM frame = a MOST frame.
J2 UART_TX: the raw control blocks (the same M2 reads).
J3: I2S, BCLK 64 Fs, 32-bit slots, 24 bits with the head unit's controls applied; the same on S/PDIF, J6, J7 (block 9).
J2 and J3 take no input: the one host is M2 (block 8).
R18..R21. R34..R41 (33 ohm): each by its FPGA pin (source termination): R38..R41 under M2.
MCLK on J2 and J3 from separate FPGA pins, no branch.
v1.2: J2's and J3's pin order follows their pins on M1's right row: the tracks do not cross.

9. DIGITAL AUDIO J5 TOSLINK IN -> U12 DIR9001 -> U13 SRC4192 -> I2S_DIN2_F; SPDIF_F -> J6 TOSLINK OUT and J7 coax

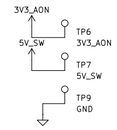


v1.3: RDY -> MUTE - silent until the SRC4192 has found the rate ratio.

J5 TOSLINK IN -> R25 -> U12 DIR9001: its PLL recovers the clock from the signal, output I2S 24 bit (FMT1 FMT0 = HH), CKSEL = L. XTI to ground: no crystal needed.
U13 SRC4192: both ports slave (MODE = 000): input I2S 24 bit (IFMT = 001) from U12; output I2S 24 bit (OFMT = 01, OWL = 00) on the FPGA's clocks (their own copies of BCLK, LRCK: SRC_BCLK_F, SRC_LRCK_F) -> I2S_DIN2_F.
X1 24.576 MHz: the reference clock of the SRC4192's rate estimator (RCKI), asynchronous to both ports. RST_AIN: R30/C40 = 10 ms, once X1 runs (the SRC4192 has no POR).
OPT_ERR (U12 ERROR) -> M2 IO13: 1 = no TOSLINK signal or a parity error; OPT_NPCM (U12 AUDIO) -> M2 IO27: 1 = not PCM (the DIR9001 does not mute AC-3/DTS): the host then does not inject. Source: command 02, bit M[2].
S/PDIF OUT: SPDIF_F (FPGA 42) -> R28 -> J6 FCR684208T: R26/R27: 0.5 V into 75 ohm, 75 ohm out; C29 blocks DC -> J7 RCA. The same stream as on J3.
U14 LP5907: 3V3_DA (250 mA) for block 9 from 5V_SW, not from the module's LDO. J7 pad 3 (the thin pin off the axis): BKL's drawing gives it no function: measure on a sample.

C31 J6: C32/C33 U12 VDD: C34/C35 U12 VCC:
C36/C37 U13 VDD: C38 U13 VIO: C39 X1: by the pins.

10. TEST POINTS (bench)



Bench probe points: the 3V3_AON and 5V_SW rails, ground. None on the ring's lines: a stub on their edges would hurt. RX_BUF at U1 pin 4, TX_SEL at U2 pin 6, TX_REGEN and RT_CLK at U9 and at M1 (TX_FB).

Repeater with regeneration and a bypass watchdog; FPGA; ESP32 host; TDM, UART, I2S out; S/PDIF in
Generated by listener_hw2/gen_listener2.py from listener_hw/gen_listener.py: the circuit of v1.6

Sheet: /
File: most_listener_v2.kicad_sch

Title: MOST25 to AUDIO: listener board v2.0

Size: A2	Date: 2026-10-11	Rev: 2.0
KiCad E.D.A. 10.0.6		Id: 1/1